Date: / /

Tutorial 1

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1) Given :

$P_1 = 0.30$, $P_2 = 0.25$, $P_3 = 0.20$, $P_4 = 0.10$, $P_5 = 0.15$, $r = 5$

$$H(x) = \sum_{i=1}^{N} P_i \log_2 \frac{1}{P_i}$$

$$= \left(0.30 \log_2 \frac{1}{0.30}\right) + \left(0.25 \log_2 \frac{1}{0.25}\right) + \left(0.20 \log_2 \frac{1}{0.20}\right) + \left(0.10 \log_2 \left(\frac{1}{0.10}\right)\right)$$

$$= (0.30)(1.737) + (0.25)(2) + (0.20)(2.32) + (0.10)(3.3219) + (0.15)(2.7369)$$

$H(x) =$ ~~1.81769~~ 2.225 bits / symbol

$R = rH = (5)(2.225) = 11.125$ bits / second

2) To prove entropy of a discrete source is max when the o/p symbols are equally probable

Consider two probabilities $P_1 = 1/2$ & $P_2 = 1/2$

$$H(x) = \sum_{i=0}^{N} P_i \log_2 \left(\frac{1}{P_i}\right)$$

$$= \frac{1}{2} \log_2 \left(\frac{1}{1/2}\right) + \frac{1}{2} \log_2 \left(\frac{1}{1/2}\right)$$

$= 1$ bit / symbol

Hence proved that entropy is maximum when the probality of two symbols is equiprobable